A2 · Unit 5 · WPH15 Thermodynamics, Radiation, Oscillations & Cosmology

Nuclear Decay

The nucleus holds enormous amounts of energy, and this page is about how it gets out. You will see why a nucleus weighs less than the parts it is made from, why both splitting heavy nuclei and joining light ones release energy, what alpha, beta and gamma radiation are, and why a lump of radioactive material decays in such a predictable way when each nucleus inside it behaves completely at random.

SPEC 133 Binding energy and mass deficit

WHAT THIS SPEC POINT ASKS

Understand nuclear binding energy, and use ΔE = c²Δm to calculate mass deficits and the energy released in nuclear processes.

Mass deficit

- A nucleus has **less mass** than the protons and neutrons (nucleons) that make it, measured separately. The difference is the **mass deficit** (mass defect) Δm:

Mass deficit

$$\Delta m = Z m_{\mathrm{p}} + (A - Z) m_{\mathrm{n}} - m_{\mathrm{nucleus}}$$

Z: proton number A: nucleon number $m_{\mathrm{p}}, m_{\mathrm{n}}$: proton and neutron masses

- Z = proton number; A = nucleon number; m_p, m_n = proton and neutron masses.

Binding energy

- The **binding energy** of a nucleus is the energy needed to **separate it completely into its individual nucleons**. Equivalently, it is the energy released when the nucleus forms from separate nucleons.
- Mass and energy are equivalent (Einstein):

Mass–energy equivalence FORMULA SHEET

$$\Delta E = c^2 \, \Delta m$$

ΔE: energy / J c: 3.00 × 10⁸ m s⁻¹ Δm: mass / kg

- ΔE = energy (J); c = 3.00 × 10⁸ m s⁻¹; Δm = mass (kg).
- Binding energy is **not** energy stored inside the nucleus. A nucleus with a large binding energy is **more** tightly bound: you'd have to put more energy in to pull it apart.
- **Binding energy per nucleon** = total binding energy ÷ A. The higher it is, the more stable the nucleus (spec point 135).

Energy released in a decay or reaction

- In any spontaneous decay, fission or fusion, the **products have less total mass** than the starting particles. The missing mass appears as energy (mostly KE of the products, plus gamma photons):

Energy released in a decay or reaction

$$\text{energy released} = c^2\,(\text{mass before} - \text{mass after})$$

The calculation chain (each step is a mark)

1. Find Δm in **u** (total mass before − total mass after; or nucleons − nucleus).
2. Convert to kg: × 1.66 × 10^{-27} kg u^{-1}.
3. ΔE = c^2Δm in J.
4. Convert to MeV: ÷ 1.60 × 10^{-13} J MeV^{-1} (or to eV: ÷ 1.60 × 10^{-19}).
5. For binding energy per nucleon, divide by A.

Keep at least 5–6 significant figures in the masses until after the subtraction: the difference is tiny.

WORKED EXAMPLE

Binding energy per nucleon of iron-56

Fe-56 has 26 protons and 30 neutrons. Mass of the nucleus = 55.92067 u; m_p = 1.00728 u; m_n = 1.00867 u.

Δm = 26 × 1.00728 + 30 × 1.00867 − 55.92067 = 26.18928 + 30.26010 − 55.92067 = 0.52871 u

Δm = 0.52871 × 1.66 × 10^{-27} = 8.777 × 10^{-28} kg

ΔE = c^2Δm = (3.00 × 10^8)² × 8.777 × 10^{-28} = 7.90 × 10^{-11} J

In MeV: 7.90 × 10^{-11} / 1.60 × 10^{-13} = 494 MeV

Per nucleon: 494/56 = **8.8 MeV**

WORKED EXAMPLE

Energy released in alpha decay

^{226}Ra → ^{222}Rn + ^{4}He. Atomic masses: Ra-226 = 226.025410 u, Rn-222 = 222.017578 u, He-4 = 4.002603 u.

Δm = 226.025410 − (222.017578 + 4.002603) = 0.005229 u

= 0.005229 × 1.66 × 10^{-27} = 8.680 × 10^{-30} kg

ΔE = (3.00 × 10^8)² × 8.680 × 10^{-30} = 7.81 × 10^{-13} J = **4.88 MeV**

Atomic masses work here because the electrons balance on both sides (88 = 86 + 2).

WORKED EXAMPLE

A helium-4 nucleus has a mass of 4.00151 u. Calculate (a) its mass deficit in kg, (b) its binding energy in MeV, and (c) its binding energy per nucleon. (proton mass = 1.00728 u, neutron mass = 1.00867 u)

Show solution

Total mass of 2 protons and 2 neutrons:

$$2(1.00728) + 2(1.00867) = 4.03190 \text{ u}$$

(a) $\Delta m = 4.03190 - 4.00151 = 0.03039 \text{ u} = 0.03039 \times 1.66 \times 10^{-27} = 5.04 \times 10^{-29} \text{ kg}$

(b) $\Delta E = c^2 \Delta m = (3.00 \times 10^8)^2 \times 5.045 \times 10^{-29} = 4.54 \times 10^{-12}$ J

$$\frac{4.54 \times 10^{-12}}{1.60 \times 10^{-13}} = 28.4 \text{ MeV}$$

(c) There are 4 nucleons, so the binding energy per nucleon is $28.4/4 = 7.1$ MeV.

EXAM TIP

Write down the mass difference in u **before** converting. Examiners give the subtraction its own mark, and "a bald answer scores zero": you need at least two of the steps written (WPH15 Oct 2021). Don't round in the middle: rounding Δm early turned 7.88 MeV into 7.85 MeV and lost the mark (WPH15 Jan 2026).

MARK-SCHEME WORDING

- "Use of ΔE = c²Δm", "conversion of u to kg" (× 1.66 × 10^{-27}), "conversion of J to MeV".
- "Do not accept use of 1.67 × 10^{-27}" for u (WPH15 Jan 2026); use the booklet value 1.66 × 10^{-27} kg.
- Binding energy per nucleon: give at least 2 s.f., worked from the masses given in the question without early rounding; an answer that drifts from the data value loses the final mark (WPH15 Oct 2025).
- Binding energy: "energy needed to separate a nucleus into its (individual) nucleons".

Often asked as: energy released in an alpha or beta decay from atomic masses; binding energy (per nucleon) from nucleon masses; mass converted to energy per second in the Sun; MCQ on what binding energy means.

COMMON MISTAKE

Applying E = mc² to the whole mass of a nucleus instead of the mass **difference**. Subtracting in the wrong order, or leaving out one of the three masses. Multiplying nuclide masses by the number of nucleons. Forgetting the 10^{-27}. Using the proton mass instead of u to convert. Dividing when you should multiply (J → MeV is ÷ 1.60 × 10^{-13}). Forgetting to divide by A for "per nucleon". Giving a decay energy when the binding energy of a nucleus was asked for.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Beta decay with atomic masses**: in β^- decay the daughter atom has one more electron than the parent's nucleus needs, so atomic masses already include the emitted electron's mass. Check what the question gives: one paper subtracted the electron mass (0.00055 u) explicitly (WPH15 Jan 2021).
- Masses in unfamiliar units such as **GeV/c²**: multiply by 10^9 × 1.60 × 10^{-19} to get J directly (WPH15 Jun 2021).
- The energy released is shared between the products by **momentum conservation** (see 139).

SPEC 134 The atomic mass unit

WHAT THIS SPEC POINT ASKS

Express small masses in atomic mass units (u) and convert between u and kg.

Definition and conversions

- The **atomic mass unit** u is **one twelfth of the mass of a carbon-12 atom**. It is roughly the mass of one nucleon.

The atomic mass unit

$$1\ \text{u} = 1.66 \times 10^{-27}\ \text{kg}$$

data booklet value

- The mass of a nucleus in u is close to (slightly less than) its nucleon number A. So a quick estimate of the mass of one nucleus is A × 1.66 × 10^{-27} kg.
- Energy equivalent of 1 u: E = $c^2\Delta m$ = $(3.00 \times 10^8)^2 \times 1.66 \times 10^{-27}$ = 1.494×10^{-10} J = **934 MeV** with booklet values (the precise value is 931.5 MeV). You may use "1 u ≈ 934 MeV" as a shortcut if you show where it comes from.
- 1 eV = 1.60×10^{-19} J; 1 MeV = 1.60×10^{-13} J.

WORKED EXAMPLE

Mass of nuclei from a count

A sample contains 3.0×10^{21} uranium-238 nuclei. Find the mass of uranium.

Mass of one nucleus ≈ $238 \times 1.66 \times 10^{-27} = 3.95 \times 10^{-25}$ kg

Total = $3.0 \times 10^{21} \times 3.95 \times 10^{-25}$ = **1.2×10^{-3} kg** (1.2 g)

MARK-SCHEME WORDING

- "× 1.66 × 10^{-27} kg u^{-1}" is the conversion mark in every decay-energy question.
- "Mass of one nucleus = A × u" for an approximate mass; "use of 1 u = 934 MeV (ecf)" accepted (WPH15 Jan 2022).

Often asked as: the conversion step inside 133; mass of a sample from the number of nuclei (often from N = A/λ in 142).

COMMON MISTAKE

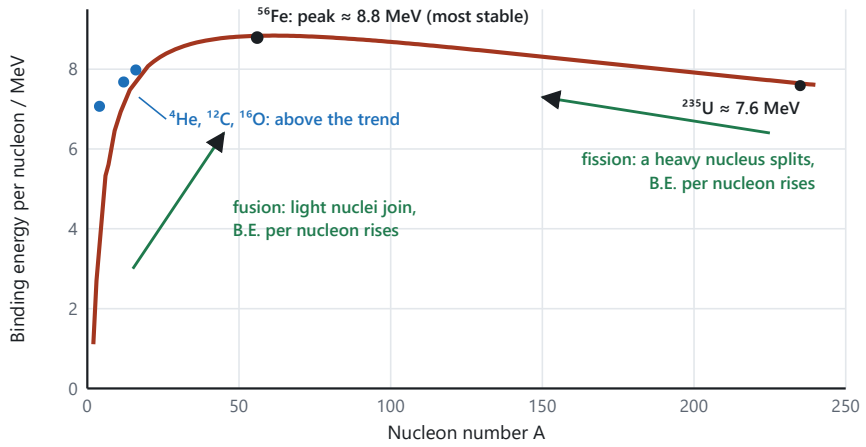
Using 1.67 × 10^{-27} kg (the proton mass) instead of 1.66 × 10^{-27} kg. Using A × m_p as the mass of a nucleus.

SPEC 135 Fission, fusion and the binding energy per nucleon curve

WHAT THIS SPEC POINT ASKS

Explain nuclear fission and fusion using the graph of binding energy per nucleon against nucleon number.

The curve



Binding energy per nucleon against nucleon number A. It rises steeply for light nuclei, peaks at about 8.8 MeV near iron-56, then falls slowly to about 7.6 MeV for uranium. He-4, C-12 and O-16 lie above the general trend.

- **Iron-56** (and its neighbours) has the **highest binding energy per nucleon**: it is the most stable nucleus.
- Any change that **moves nuclei up the curve** (towards iron) **releases energy**, because the products are more tightly bound and so have **less total mass**.

Fusion

- **Fusion**: two **light** nuclei join to make a heavier nucleus. For light nuclei (left of the peak) the binding energy per nucleon **increases**, so energy is released.
- The curve is steepest here, so fusion releases more energy **per kilogram of fuel** than fission.

Fission

- **Fission**: a **heavy** nucleus (e.g. U-235, after absorbing a slow neutron) **splits** into two medium-sized nuclei plus 2 or 3 neutrons. For nuclei right of the peak, the products have a **higher binding energy per nucleon**, so energy is released.
- The released neutrons can cause further fissions: a **chain reaction**.
- Fission of a nucleus **lighter** than iron (e.g. He-4) would **absorb** energy, because the products would have a lower binding energy per nucleon.

Energy released from graph readings

Energy released from the curve

$$\text{energy released} = (\text{total binding energy of products}) - (\text{total binding energy of reactants})$$

with total binding energy = A × (binding energy per nucleon) for each nucleus.

WORKED EXAMPLE

Energy from fusion and fission using the curve

Fusion $^{2}H + {}^{3}H \rightarrow {}^{4}He + n$. Binding energies per nucleon: ^{2}H 1.11 MeV, ^{3}H 2.83 MeV, ^{4}He 7.07 MeV (a lone neutron has none).

Reactants: 2 × 1.11 + 3 × 2.83 = 10.71 MeV

Products: 4 × 7.07 = 28.28 MeV

Released = 28.28 − 10.71 = **17.6 MeV**

Fission ^{235}U + n → ^{141}Ba + ^{92}Kr + 3n. Readings: U-235 7.6 MeV, Ba-141 8.3 MeV, Kr-92 8.5 MeV.

Released = (141 × 8.3 + 92 × 8.5) − 235 × 7.6 = 1952 − 1786 = **about 170 MeV**

Per nucleon of fuel: fusion 17.6/5 ≈ 3.5 MeV; fission 170/236 ≈ 0.7 MeV. Fusion gives more energy per kilogram.

WORKED EXAMPLE

In a nuclear reactor, uranium-235 absorbs a neutron and splits: $^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n}$. The binding energies per nucleon are U-235: 7.59 MeV, Ba-141: 8.33 MeV, Kr-92: 8.51 MeV. (a) Calculate the energy released in one fission. (b) Estimate the energy released if 1.0 kg of uranium-235 undergoes fission.

Show solution

(a) Free neutrons have no binding energy, so only the nuclei count.

$$\text{before: } 235 \times 7.59 = 1784 \text{ MeV}$$

$$\text{after: } 141 \times 8.33 + 92 \times 8.51 = 1174.5 + 782.9 = 1957 \text{ MeV}$$

$$\text{energy released} = 1957 - 1784 = 174 \text{ MeV} = 174 \times 1.60 \times 10^{-13} = 2.78 \times 10^{-11} \text{ J}$$

(More energy, bringing the total to about 200 MeV, is released later as the neutron-rich fission products decay.)

(b) Mass of one U-235 nucleus $\approx 235 \times 1.66 \times 10^{-27} = 3.90 \times 10^{-25}$ kg, so 1.0 kg contains

$$N = \frac{1.0}{3.90 \times 10^{-25}} = 2.56 \times 10^{24} \text{ nuclei}$$

$$E = 2.56 \times 10^{24} \times 2.78 \times 10^{-11} = 7.1 \times 10^{13} \text{ J}$$

Burning 1 kg of coal releases about 3×10^{7} J, so this is over two million times more.

EXAM TIP

To sketch the curve for marks: start near the origin and rise **steeply**, peak at **Fe-56** (label it, about 8–9 MeV), then fall **slowly**. Label the y-axis "binding energy **per nucleon** / MeV". Read a total binding energy from the graph as A × (the y-value at that A): the superscript in the symbol is the A you need.

MARK-SCHEME WORDING

- Fission: "a large nucleus splits (into two (smaller) nuclei and neutrons)"; "binding energy (per nucleon) increases so energy is released", or "decrease in (total) mass so energy is released" (WPH15 Jan 2024).
- Fusion: "light nuclei combine"; "binding energy per nucleon increases".
- Graph sketch: "steeply rising curve near to origin; slowly decreasing curve after peak; iron-56 marked at peak" (WPH15 Jun 2021).
- Fe-56 has the greatest binding energy **per nucleon**, not the greatest total binding energy (WPH15 Jan 2023).

Often asked as: define fission; sketch the curve and mark Fe-56; balance a fission equation; explain how fission or fusion releases energy; energy released from graph readings; compare Fe and U binding energies.

COMMON MISTAKE

"Binding energy decreases in fission, so energy is released" is wrong: it **increases**. Using the word "fission" without saying what happens. Confusing total and per-nucleon binding energy. Only counting one fission product's binding energy.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Balancing a fission equation including the neutrons: $^{236}U \rightarrow {}^{93}Sr + {}^{141}Xe + 2n$ (WPH15 Jun 2021).
- A trap: fission of He-4 would **decrease** binding energy per nucleon, so it would not release energy (WPH15 Oct 2025).
- Estimating U-235's binding energy per nucleon from the graph by proportion (about 7.4–7.6 MeV) (WPH15 Oct 2025).

SPEC 136 Conditions for nuclear fusion

WHAT THIS SPEC POINT ASKS

Understand why fusion needs very high temperatures and very high densities to start and to keep going.

Why it is hard

- Nuclei are **positively charged**, so they **repel** each other electrostatically. The repulsion grows as they get closer.
- The **strong nuclear force** that can bind them together only acts over a **very short range** (about 10^{-15} m, the size of a nucleus).
- To fuse, two nuclei must get **close enough for the strong force to act**. They must approach with a very large kinetic energy to overcome the repulsion.

The two conditions

1. **Very high temperature** (about 10^7 K in the Sun's core; around 10^8 K in Earth reactors): mean KE $\propto$ T, so the nuclei move fast enough to overcome the electrostatic repulsion. At these temperatures matter is a **plasma**: electrons are stripped from the nuclei.
2. **Very high density** (a large number of nuclei per unit volume): collisions happen **often enough** to keep the fusion rate high, so enough energy is released to **maintain** the high temperature and sustain fusion.

Where these happen

- In a **star**, gravity squeezes the core to a huge density, and the gravitational collapse raises the temperature.
- On **Earth**, the plasma must be **confined** by strong **magnetic fields**: it can't touch a solid wall, which would cool it and stop fusion (and would be damaged). Keeping the plasma hot, dense and confined for long enough is why fusion power is so difficult.
- In the Sun, hydrogen fuses to helium (net: 4 protons → 1 helium-4 nucleus + 2 positrons + 2 neutrinos + energy).

WORKED EXAMPLE

The Sun's mass loss

The Sun's luminosity is 3.85×10^{26} W. How much mass does it convert to energy each second?

$\Delta m/\Delta t = L/c^2 = 3.85 \times 10^{26} / (3.00 \times 10^8)^2 =$ **4.3×10^9 kg s^{-1}**

WORKED EXAMPLE

Estimate the temperature needed for two deuterium nuclei to get within 3.0 fm of each other, assuming the energy available in a collision is about $\frac{3}{2}kT$.

Show solution

The energy needed is the electrostatic potential energy at 3.0 fm (each deuterium nucleus has charge $+e$):

$$E_{\mathrm{p}} = \frac{8.99 \times 10^{9} \times (1.60 \times 10^{-19})^{2}}{3.0 \times 10^{-15}} = 7.67 \times 10^{-14}\ \mathrm{J}$$

Set $\frac{3}{2}kT = E_{\mathrm{p}}$:

$$T = \frac{2E_{\mathrm{p}}}{3k} = \frac{2 \times 7.67 \times 10^{-14}}{3 \times 1.38 \times 10^{-23}} = 3.7 \times 10^{9}\ \mathrm{K}$$

This is only an estimate, but it shows that temperatures of billions of kelvin are needed if every collision has just the average energy.

MARK-SCHEME WORDING

The indicative content is the same in every paper (WPH15 Oct 2021, Jan 2024, Oct 2024, Jan 2026, Jun 2026):

- (very) **high temperature**;
- so the nuclei/protons have enough **kinetic energy** ("kinetic energy, not just energy");
- to **overcome the electrostatic repulsion** between the positively charged nuclei;
- so the nuclei get **close enough to fuse** (within range of the strong force);
- (very) **high density**;
- so there is a **high collision rate**, to sustain fusion / maintain the temperature.

"Not 'particles', not 'molecules', not 'atoms'": say **nuclei** or **protons**. Accept "collide more often"; reject "number of collisions" (WPH15 Jun 2026).

Often asked as: a 4-mark short answer or a 6-mark linkage question on fusion in a star's core or a reactor; why Earth fusion needs magnetic confinement; mass lost per second by the Sun.

COMMON MISTAKE

Writing "particles" or "atoms" instead of nuclei. Saying high density means "more collisions" without "per second" (rate). Answering about the balance of forces inside a star (that's a different question). Saying "energy" when the mark needs "kinetic energy".

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Plasma containment on Earth with strong magnetic fields (WPH15 Oct 2020).
- Mass converted per second from L/c², and the fraction of the Sun's mass used up over a time (WPH15 Oct 2020, Jun 2024).

SPEC 137 Background radiation

WHAT THIS SPEC POINT ASKS

Know that background radiation is always present and correct measured count rates for it.

Sources

- **Natural**: **radon** gas seeping from rocks and soil (the largest share in many places; an alpha emitter, hazardous when breathed in); rocks and buildings (granite); **cosmic rays** from space; food and drink (potassium-40, carbon-14), so our own bodies.
- **Artificial**: medical X-rays and tracers (the biggest artificial share); fallout from past weapons tests and accidents; nuclear industry (very small).
- It varies with location (rock type, altitude).

Correcting for it

1. With **no source present**, measure the count over a **long time** (several minutes) and work out the background count rate.
2. **Subtract** it from **every** reading taken with the source:

Background correction

$$\text{corrected count rate} = \text{measured count rate} - \text{background count rate}$$

1. Do the subtraction **before** any further processing (logs, ratios, half-lives).

- A decay curve that levels off **above zero**, or a count that never falls below a steady value, is showing the background.
- Radioactive counts fluctuate randomly, so longer counting times give more reliable averages.

WORKED EXAMPLE

Corrected count rate

Background: 270 counts in 10 minutes. With the source: 1950 counts in 5 minutes. Find the corrected count rate.

Background rate = 270/10 = 27 counts per minute

Measured rate = 1950/5 = 390 counts per minute

Corrected = 390 − 27 = **363 counts per minute**

WORKED EXAMPLE

With no source present, a GM tube records 186 counts in 10 minutes. A source is then placed in front of it and the tube records 1240 counts in 2 minutes. Calculate the corrected count rate in counts per second.

Show solution

Background count rate $= 186/600 \text{ s} = 0.31 \text{ s}^{-1}$.

Measured count rate $= 1240/120 \text{ s} = 10.33 \text{ s}^{-1}$.

Corrected count rate $= 10.33 - 0.31 = 10.0 \text{ s}^{-1}$.

MARK-SCHEME WORDING

- "Measure the count (rate) with no source present (over a long time)"; "subtract (the background count rate) from each reading".
- Without the correction "the count rate is overestimated" / "background count introduces a **systematic** error". Just saying "it gives a more accurate value" scored nothing (WPH15 Jun 2024).

Often asked as: MCQs on how background affects a measurement (WPH15 Jun 2021, Jan 2026); deduce whether an activity has fallen to background level; describe how to find the background and why it causes a systematic error.

COMMON MISTAKE

Subtracting the background from the **initial** activity in a calculation where the question's numbers are already corrected. Muddling initial and final activities when finding the time to fall to background.

SPEC 138 Alpha, beta and gamma radiation

WHAT THIS SPEC POINT ASKS

Relate the nature of alpha, beta and gamma radiation to their ionising ability, penetration and range in different materials.

Properties

	Alpha (α)	Beta-minus (β^-)	Gamma (γ)
Nature	helium nucleus: 2 protons + 2 neutrons	fast **electron** from the nucleus	high-energy **electromagnetic** wave (photon)
Charge	**+2e**	**−e**	0
Mass	about 4 u	about 1/1840 u	0
Ionising ability	**strong**	moderate	weak
Range in air	a few cm	tens of cm to about 1 m	very long (intensity falls with distance)
Stopped by	a sheet of paper, or skin	a few mm of aluminium	reduced (never fully stopped) by several cm of lead or metres of concrete
Deflection in fields	small (heavy), towards the negative plate	large (light), opposite way to α	not deflected

(β^+ is a **positron**, charge **+e**; its properties are otherwise like β^-, but it annihilates with an electron almost at once.)

Ionising ability links to range

- **Ionisation**: the radiation knocks electrons out of atoms it passes, losing some energy each time.
- **Alpha** is massive and doubly charged, so it interacts strongly and ionises densely. It loses its energy fast, so it has a **short range**: **the more strongly ionising, the shorter the range**.
- **Gamma** is uncharged and interacts rarely, so it ionises weakly and penetrates far. Its intensity falls **exponentially** with absorber thickness (spec point 140), so there is no definite range.

Safety

- Outside the body, alpha is the **least** dangerous (skin stops it). **Inside** the body (inhaled or swallowed) it is the **most** dangerous, because all its ionisation is done in a small volume of living tissue.
- Handle sources with **tongs** (distance), for as **short a time** as possible ("the ionising effect is cumulative"), with **shielding**; store in lead-lined boxes.

MARK-SCHEME WORDING

- Cloud chamber: "the thickness of the track is related to the ionising ability: alpha is strongly ionising and beta only moderately ionising (so alpha tracks are thick, beta tracks thin)"; "the shape of the track is related to the mass: alpha particles are massive (so straight tracks) and beta are not (so twisted tracks)" (WPH15 Jun 2023).
- Alpha: "can only travel a few cm in air", "only a danger if taken into the body" (WPH15 Jun 2026).
- Safety: "use tongs", "minimise time of exposure as the ionising effect is cumulative" (WPH15 Oct 2020).

Often asked as: comment on each radiation in turn in a given context; cloud-chamber tracks; safety with a source; shielding thickness from a graph; tracer choice.

COMMON MISTAKE

GCSE-style lists of materials (lead, concrete) that ignore the context in the question. Not commenting on each radiation separately. Confusing "ionisation energy" with ionising ability. Linking a twisting track to speed rather than to the particle's small **mass**. Giving beta a charge of +1e: β^- is −e (only the positron, β^+, is +e).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **PET scans**: a β^+ tracer emits positrons that **annihilate** with electrons, producing two gamma photons that pass out of the body to the detectors. The tracer's half-life must be long enough to do the scan but short enough to limit the dose (WPH15 Oct 2022).
- An alpha particle ionising air excites molecules, which emit **UV photons** as they de-excite; the weakly ionising UV travels much further, so it can be detected at a safe distance (WPH15 Jun 2021).
- The fraction of beta absorbed rises with paper thickness: quote the relationship, not vague "absorbs more" (WPH15 Jan 2025).

SPEC 139 Nuclear equations

WHAT THIS SPEC POINT ASKS

Write and interpret nuclear equations for decays and reactions, given the particle symbols.

Rules

- Write each nucleus as ${}^{A}_{Z}X$ (A = nucleon number on top, Z = proton number below).
- **Nucleon number (top) is conserved** and **charge/proton number (bottom) is conserved**: the totals are the same on both sides.

The particle symbols

Particle	Symbol	Top (A)	Bottom (Z)
alpha	${}^{4}_{2}\alpha$ or ${}^{4}_{2}\mathrm{He}$	4	2
beta-minus (electron)	${}^{0}_{-1}\beta$ or ${}^{0}_{-1}\mathrm{e}$	0	−1
beta-plus (positron)	${}^{0}_{+1}\beta$ or ${}^{0}_{+1}\mathrm{e}$	0	+1
neutron	${}^{1}_{0}\mathrm{n}$	1	0
proton	${}^{1}_{1}\mathrm{p}$	1	1

Particle	Symbol	Top (A)	Bottom (Z)
gamma	$^{0}_{0}\gamma$	0	0
electron antineutrino / neutrino	$\bar{\nu}_e$ / ν_e	0	0

The decays

- **Alpha**: A falls by 4, Z falls by 2.
  $^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\alpha$
- **Beta-minus**: a neutron becomes a proton ($n \rightarrow p + e^- + \bar{\nu}_e$). A unchanged, Z **rises** by 1. An **antineutrino** is emitted too.
  $^{90}_{38}\text{Sr} \rightarrow {}^{90}_{39}\text{Y} + {}^{0}_{-1}\beta + \bar{\nu}_e$
- **Beta-plus**: a proton becomes a neutron ($p \rightarrow n + e^+ + \nu_e$). A unchanged, Z **falls** by 1. A **neutrino** is emitted.
  $^{18}_{9}\text{F} \rightarrow {}^{18}_{8}\text{O} + {}^{0}_{+1}\beta + \nu_e$
- **Gamma**: the nucleus drops from an excited state to a lower energy state. A and Z don't change. It often follows alpha or beta decay.
- **Fission** and **fusion** obey the same rules: $^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3\,{}^{1}_{0}\text{n}$ (top: 236 = 141 + 92 + 3; bottom: 92 = 56 + 36).

How the energy is shared

- **Alpha decay** (two products): momentum is conserved, so the alpha and the daughter nucleus move off with **equal and opposite momentum**. Since $E_k = p^2/2m$, the much lighter alpha gets almost all the KE, and every alpha from that decay has the **same energy**.
- **Beta decay** (three products: daughter, electron, antineutrino): the energy can be shared in many ways, so beta particles have a **continuous range of energies** up to a maximum. This was the evidence for the neutrino.

WORKED EXAMPLE

KE of the alpha particle

In the Ra-226 decay 4.88 MeV is released. The alpha (4 u) and radon (222 u) share it. Find the alpha's KE.

Equal momentum p, and $E_k = p^2/2m$, so $E_k \propto 1/m$.

$E_\alpha = 4.88 \times 222/(222 + 4) =$ **4.79 MeV** (the radon gets only 0.09 MeV)

WORKED EXAMPLE

Uranium-238 ($Z = 92$) decays by emitting an alpha particle, and the product then emits two beta-minus particles. (a) Identify the final nucleus. (b) Uranium-238 eventually becomes stable lead-206 ($Z = 82$). How many alpha and beta-minus decays happen along the way?

Show solution

(a) Alpha: $A = 238 - 4 = 234$, $Z = 92 - 2 = 90$ (thorium-234). Each beta-minus adds 1 to Z: $90 \rightarrow 91 \rightarrow 92$. The final nucleus is $^{234}_{92}\text{U}$, a different isotope of uranium.

(b) Only alpha decay changes A, by 4 each time: $238 - 206 = 32$, so there are $32/4 = 8$ alpha decays. Those would reduce Z by 16, from 92 to 76. To end at 82, Z must rise by 6, so there are **6 beta-minus decays**.

MARK-SCHEME WORDING

- Top line (nucleon numbers) and bottom line (proton numbers) are separate marks.
- Beta-minus: electron as $^{0}_{-1}\text{e}$ "plus an antineutrino"; the daughter has **Z + 1**.

- Range of beta energies: "momentum/KE is **shared between three** particles"; "the energy split between the beta particle and the (anti)neutrino is random" (WPH15 Jan 2021).
- Alpha: "momentum is conserved, so the velocity of the heavy nucleus is much less, so its KE is much smaller than that of the alpha" (WPH15 Oct 2021).

Often asked as: complete a β^- equation with the antineutrino (Sr-90, tritium, K-40); balance alpha, fission or fusion equations; why beta energies vary; why the recoiling nucleus gets almost no KE.

COMMON MISTAKE

Writing the beta particle as A = −1, Z = 0 (upside down). Subtracting 1 from Z in β^- decay, or combining −1 and 38 to give 37 for Y-90 (it's 39). Forgetting the antineutrino. Leaving out the neutrons in a fission equation.

SPEC 140 Core practical 15: absorption of gamma radiation by lead

WHAT THIS SPEC POINT ASKS

Investigate how the count rate of gamma radiation depends on the thickness of lead absorber.

The relationship

- Gamma intensity (and corrected count rate R) falls **exponentially** with absorber thickness x:

Absorption of gamma radiation

$$R = R_0\, e^{-\mu x}$$

R_0: corrected count rate with no absorber μ: attenuation coefficient / cm^{-1} or m^{-1} x: absorber thickness

- R_0 = corrected count rate with no absorber; μ = **attenuation (absorption) coefficient** (unit: cm^{-1} or m^{-1}, the inverse of x's unit).
- Take natural logs to get a straight line:

Straight-line form

$$\ln R = \ln R_0 - \mu x$$

gradient of $\ln R$ against x is $-\mu$

- Plot ln R (y) against x: **gradient = −μ**, y-intercept = $\ln R_0$.
- **Half-value thickness** (thickness that halves the count rate): $x_{½} = \ln 2/\mu$.

CORE PRACTICAL

Core practical 15: method

1. With no source nearby, record the **background** count over at least 5 minutes; find the background count rate.
2. Clamp the gamma source and the Geiger–Müller tube a fixed distance apart (e.g. 10 cm). Record the count over a fixed time (e.g. 1 minute) with no lead.
3. Measure each lead sheet's thickness with a **micrometer** (or vernier calipers) at several points and average.
4. Put lead sheets between source and tube, building up the thickness. For each thickness take **three** 1-minute counts and average.

5. Subtract the background from every count rate, then plot ln(corrected count rate) against thickness.

Variables: independent = lead thickness; dependent = count rate; control = source–detector distance, same source and detector, counting time, position (same background).

Safety: lift the source with **long tongs**, point it away from people, keep it at arm's length and return it to its lead-lined box straight away; minimise time near it; wash hands after handling lead.

Improvements: count for longer to reduce the effect of random fluctuations; keep the distance fixed with a clamp and ruler.

WORKED EXAMPLE

Attenuation coefficient and half-value thickness

Corrected count rates (counts per minute): x = 0 cm, 480; 0.5 cm, 374; 1.0 cm, 291; 1.5 cm, 227; 2.0 cm, 177; 2.5 cm, 138.

ln R: 6.17, 5.92, 5.67, 5.42, 5.17, 4.92 (a straight line, so the decay is exponential).

Gradient (large triangle, ends of the line) = (4.92 − 6.17)/(2.5 − 0) = −0.50 cm^{-1}, so **$\mu = 0.50\ cm^{-1}$**

$x_{½}$ = ln 2/μ = 0.693/0.50 = **1.4 cm**

Check: R at 1.4 cm ≈ 480 × $e^{-0.70}$ ≈ 240, half of 480.

MARK-SCHEME WORDING

- "Log expansion of $R = R_0e^{-\mu x}$", "μ identified as (−) gradient", "gradient calculated", "large triangle used", "conclusion consistent with the half-value thickness" (WPH15 Jan 2021).
- Give μ **with its unit** (cm^{-1} or m^{-1}): many correct values lost the unit mark (WPH15 Jun 2024).
- From a curve, μ = 1/(the intercept the tangent at x = 0 makes on the x-axis) (WPH15 Jun 2024).

Often asked as: process log data to find μ and compare a half-value thickness with a claim; read μ from a graph; percentage transmission and shield thickness; background correction MCQs; safety.

COMMON MISTAKE

Taking a gradient on a log-scale plot of the raw data instead of on a ln R against x graph. Forgetting to subtract background before taking logs. Spending time on distractor data (such as the gamma energy) in the stem. Power-of-ten errors when converting cm and m.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The equation $R = R_0e^{-\mu x}$ and the term "attenuation coefficient" are not in the spec, but appear in the question stem and must be used (WPH15 Jan 2021).
- Reading values from a log scale (the y-axis printed with 10, 100, 1000…) (WPH15 Jan 2022, Jan 2025).

SPEC 141 The random and spontaneous nature of radioactive decay

WHAT THIS SPEC POINT ASKS

Understand that radioactive decay is spontaneous and random.

The two words

- **Random**: you **cannot predict which** nucleus will decay next, or **when** any particular nucleus will decay. Each nucleus has the same, constant **probability** of decaying per unit time (the decay constant λ), regardless of how long it has existed.
- **Spontaneous**: the decay is **not affected by external factors**: not temperature, pressure, chemical combination, or the presence of other nuclei. Nothing triggers it.

Consequences

- Because every nucleus has the same probability λ per second, the number decaying per second is proportional to the number present: **dN/dt = −λN** (spec point 142). That is why decay is **exponential** and the half-life is constant.
- With huge numbers of nuclei, the **average** behaviour is very predictable, even though each decay is random.
- **Evidence**: counts recorded over equal time intervals **fluctuate** about a mean; the clicks of a Geiger counter are irregular. The spread is about √N for N counts, so longer counts give smaller percentage uncertainty.

MARK-SCHEME WORDING

- Random: "it is not possible to predict **which** nucleus will decay next, or **when**".
- Spontaneous: "not affected by external factors such as temperature or pressure" (or by the presence of other nuclei).
- The false statement in a 2026 MCQ was "it is possible to predict which nucleus will decay next" (WPH15 Jun 2026).

Often asked as: MCQs testing the two definitions (only about a third answered the WPH15 Jan 2024 one correctly); used as the justification for dN/dt = −λN and for fluctuating counts.

COMMON MISTAKE

Swapping the meanings: "random" is about prediction; "spontaneous" is about outside influences. Saying the half-life of a single nucleus is predictable (only the behaviour of a large number is).

SPEC 142 Half-life and the decay equations

WHAT THIS SPEC POINT ASKS

Find half-lives graphically and use A = λN, dN/dt = −λN, λ = ln2/t½, $N = N_0e^{-\lambda t}$ and $A = A_0e^{-\lambda t}$; derive and use the corresponding log equations.

Activity and the decay constant

- **Activity** A = number of decays per second, unit **becquerel** (Bq = s^{-1}).
- **Decay constant** λ = probability that a given nucleus decays per unit time (s^{-1}, or h^{-1}, y^{-1}...).

Activity

FORMULA SHEET

$$A = \lambda N$$

A: activity / Bq λ: decay constant / s^{-1}
N: number of undecayed nuclei

Rate of decay

FORMULA SHEET

$$\frac{dN}{dt} = -\lambda N$$

dN/dt: rate of change of N / s^{-1}

- The minus sign shows N **decreases**. A is the size of dN/dt.

- For A in Bq, λ must be in **s⁻¹**.

Exponential decay

Number of undecayed nuclei

FORMULA SHEET

$$N = N_0\, e^{-\lambda t}$$

N_0: number at $t = 0$

Activity

FORMULA SHEET

$$A = A_0\, e^{-\lambda t}$$

A_0: activity at $t = 0$ / Bq

- N_0, A_0 = values at t = 0. λ and t must be in **matching** time units (λ in day⁻¹ with t in days is fine).
- The mass of the parent and the corrected count rate fall in the same way (both ∝ N).

Half-life

- **Half-life** t½: the **time taken for the number of undecayed nuclei (or the activity) to halve**. It is constant for a given isotope.
- Derivation: put N = N₀/2 at t = t½:
  $N_0/2 = N_0e^{-\lambda t½} \rightarrow ½ = e^{-\lambda t½} \rightarrow e^{\lambda t½} = 2 \rightarrow \lambda t½ = \ln 2$, so

Half-life and decay constant

FORMULA SHEET

$$t_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$$

$t_{1/2}$: half-life λ: decay constant, in the matching time unit

Where λ = ln 2 / t½ comes from

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. After one half-life, half of the nuclei are left: $N = \frac{1}{2}N_0$ when $t = t_{1/2}$. Put this into $N = N_0e^{-\lambda t}$:

$$\tfrac{1}{2}N_0 = N_0\, e^{-\lambda t_{1/2}} \quad \Rightarrow \quad \tfrac{1}{2} = e^{-\lambda t_{1/2}}$$

2. Take the natural log of both sides. $\ln \frac{1}{2} = -\ln 2$, and $\ln(e^x) = x$:

$$-\ln 2 = -\lambda t_{1/2} \quad \Rightarrow \quad \boxed{\lambda = \frac{\ln 2}{t_{1/2}}}$$

So a large decay constant means a short half-life. $\ln 2 \approx 0.693$.

- After n half-lives the fraction left is (½)ⁿ. This works for whole numbers of half-lives; for anything else (e.g. "10% remaining") use logs.

The log forms

- Take natural logs of $A = A_0e^{-\lambda t}$:

Log form

$$\ln A = \ln A_0 - \lambda t$$

gradient of $\ln A$ against t is $-\lambda$; the same works for N

Deriving the log equations: ln N = ln N₀ − λt

SPEC: YOU MUST BE able TO DERIVE THIS

1. Start from the decay equation:

$$N = N_0\, e^{-\lambda t}$$

2. Take the natural log of both sides:

$$\ln N = \ln(\, N_0\, e^{-\lambda t})$$

3. The log of a product is the sum of the logs, $\ln(ab) = \ln a + \ln b$:

$$\ln N = \ln N_0 + \ln(\, e^{-\lambda t})$$

4. $\ln$ undoes e, so $\ln(e^{-\lambda t}) = -\lambda t$:

$$\boxed{\ln N = \ln N_0 - \lambda t}$$

5. The same steps applied to $A = A_0 e^{-\lambda t}$ give the log equation for activity (and for count rate, once background has been subtracted):

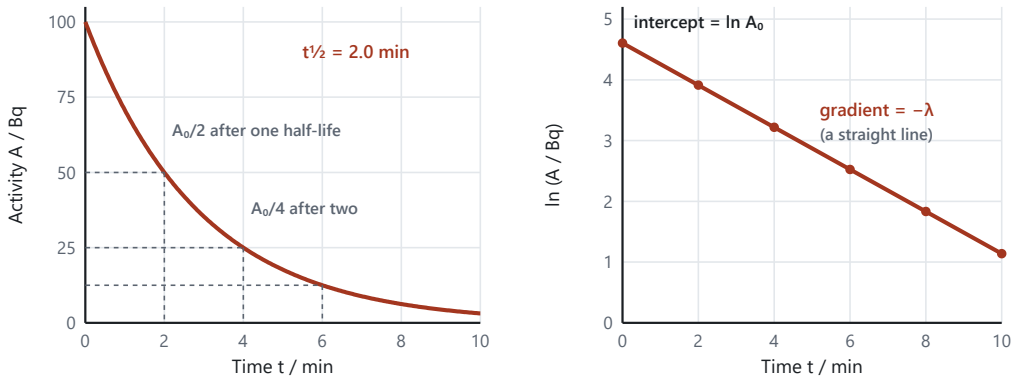
$$\boxed{\ln A = \ln A_0 - \lambda t}$$

> *Compare with $y = mx + c$: plotting $\ln N$ (or $\ln A$) on the y-axis against t gives a straight line with gradient $-\lambda$ and y-intercept $\ln N_0$.*

- Compare with y = mx + c: plotting **ln A against t** gives a straight line with **gradient −λ** and intercept ln A₀. The same works for N: ln N = ln N₀ − λt.
- Rearranged for time:

Time from activities

$$t = \frac{1}{\lambda}\ln\left(\frac{A_0}{A}\right)$$



Left: activity halves every half-life (here 2.0 min), and the curve never reaches zero. Right: ln A against t is a straight line of gradient −λ, which is the best way to find λ from data.

Finding the half-life from a graph

- **From an A–t (or N–t) curve**: read A_0, find when $A = A_0/2$; check with $A_0/4$ and $A_0/8$ (they should come at $2t_{½}$ and $3t_{½}$) and average.
- **From a ln A–t line**: gradient = −λ, then $t_{½} = \ln 2/\lambda$. Use a large triangle.
- Subtract **background** from count rates first.

Modelling decay step by step

The rate equation $\Delta N/\Delta t = -\lambda N$ can be modelled on a spreadsheet: choose a small Δt, then repeatedly calculate $\Delta N = -\lambda N \Delta t$ and $N_{new} = N + \Delta N$. The smaller Δt is, the closer the result to the exponential curve.

WORKED EXAMPLE

Iodine-131 ($t_{½}$ = 8.0 days)

A sample has activity 4.0×10^8 Bq.

(a) $\lambda = \ln 2/t_{½} = 0.693/(8.0 \times 24 \times 3600\ \text{s}) =$ **$1.0 \times 10^{-6}\ \text{s}^{-1}$**

(b) $N = A/\lambda = 4.0 \times 10^8 / 1.0 \times 10^{-6} =$ **4.0×10^{14} nuclei**

(c) Activity after 30 days ($\lambda = 0.693/8.0 = 0.0866\ \text{day}^{-1}$, so t can stay in days):

$A = 4.0 \times 10^8 \times e^{-0.0866 \times 30} = 4.0 \times 10^8 \times 0.0743 =$ **3.0×10^7 Bq**

(d) Time to fall to 1.0% of A_0:

$t = (1/\lambda)\ln(A_0/A) = (1/0.0866) \times \ln(100) = 4.61/0.0866 =$ **53 days**

WORKED EXAMPLE

Activity from a mass

Find the activity of 5.0 mg of cobalt-60 ($t_{½}$ = 5.27 years; 1 year = 3.15×10^7 s).

$N =$ mass/mass of one atom $= 5.0 \times 10^{-6}\ \text{kg} / (60 \times 1.66 \times 10^{-27}\ \text{kg}) = 5.02 \times 10^{19}$

$\lambda = 0.693/(5.27 \times 3.15 \times 10^7\ \text{s}) = 4.18 \times 10^{-9}\ \text{s}^{-1}$

$A = \lambda N = 4.18 \times 10^{-9} \times 5.02 \times 10^{19} =$ **2.1×10^{11} Bq**

WORKED EXAMPLE

Carbon dating

Living wood has a C-14 activity of 0.25 Bq per gram of carbon. Ancient wood gives 0.21 Bq per gram. t½ of C-14 = 5730 years. How old is it?

$\lambda = 0.693/5730 = 1.21 \times 10^{-4}\ y^{-1}$

$t = (1/\lambda)\ \ln(A_0/A) = \ln(0.25/0.21)/1.21 \times 10^{-4} = 0.174/1.21 \times 10^{-4} =$ **1400 years**

WORKED EXAMPLE

A hospital has a sample of iodine-131 of mass 2.0 μg. Iodine-131 has a half-life of 8.0 days. (a) Show that the sample contains about 9×10^{15} nuclei. (b) Calculate its activity. (c) Calculate its activity 30 days later.

Show solution

(a) Mass of one nucleus $\approx 131 \times 1.66 \times 10^{-27} = 2.17 \times 10^{-25}$ kg, so

$$N = \frac{2.0 \times 10^{-9}}{2.17 \times 10^{-25}} = 9.2 \times 10^{15}$$

(b) The half-life must be in seconds for an activity in Bq: $t_{1/2} = 8.0 \times 24 \times 3600 = 6.91 \times 10^{5}$ s.

$$\lambda = \frac{\ln 2}{6.91 \times 10^{5}} = 1.00 \times 10^{-6}\ \text{s}^{-1}, \qquad A = \lambda N = 1.00 \times 10^{-6} \times 9.2 \times 10^{15} = 9.2 \times 10^{9}\ \text{Bq}$$

(c) In days, $\lambda = \ln 2/8.0 = 0.0866\ \text{day}^{-1}$, so $\lambda t = 0.0866 \times 30 = 2.60$:

$$A = A_0 e^{-\lambda t} = 9.2 \times 10^{9} \times e^{-2.60} = 6.8 \times 10^{8}\ \text{Bq}$$

Notice you can use any time unit for λt, as long as λ and t use the same one.

WORKED EXAMPLE

Carbon-14 has a half-life of 5730 years. A piece of ancient wood has a corrected count rate that is 30% of the rate from the same mass of living wood. Estimate the age of the wood.

Show solution

$\lambda = \ln 2/5730 = 1.21 \times 10^{-4}\ \text{year}^{-1}$. Using the log equation with $A/A_0 = 0.30$:

$$\ln A = \ln A_0 - \lambda t \quad \Rightarrow \quad t = \frac{\ln(A_0/A)}{\lambda} = \frac{\ln(1/0.30)}{1.21 \times 10^{-4}} = 9.95 \times 10^{3}\ \text{years}$$

Check: 30% is between $\frac{1}{2}$ (one half-life) and $\frac{1}{4}$ (two half-lives), and 9950 years (about 10 000 years) is between 5730 and 11 460 years.

EXAM TIP

Decide first: does the question want t, A or N? Then choose units: if you need A = λN (so A in Bq), λ **must** be in s^{-1}; if you only use $A = A_0e^{-\lambda t}$, any matching units work. Keep 3–4 s.f. in λ, especially when A and A_0 are close, and finish with the comparison if there's a claim.

MARK-SCHEME WORDING

- "Use of $\lambda = \ln 2/t_{1/2}$", "use of $A = \lambda N$", "use of $A = A_0 e^{-\lambda t}$", "conversion between seconds and years" (its own mark) (WPH15 Oct 2025).
- "Allow ecf for λ." For deduce questions the final mark is the comment: "X > Y so not consistent", or "X ≈ Y so consistent" (WPH15 Jun 2026).
- Power from a source: P = A × (energy per decay in J) (WPH15 Jun 2021).

Often asked as: find N or N_0; time for the activity to fall to a level; activity after a time, "is the claim true?"; compare two isotopes; power from a radioactive source; percentage composition from activity; MCQ calculating λ in s^{-1}.

COMMON MISTAKE

Putting the **half-life** into the exponent instead of the elapsed time. Mixing up N and N_0, or A and A_0 (a measured "2.35 GBq" was A, not A_0). Not converting days or years to seconds before A = λN. Treating "0.75%" as 0.75 instead of 0.0075. Using $(\frac{1}{2})^n$ with a non-whole n without logs. Concluding "the claim is accurate" when your number disagrees, even at 1 s.f.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Number of nuclei from a sample's mass and the atomic mass unit (WPH15 Oct 2025, Jun 2026).
- When the time is much shorter than the half-life, the activity is nearly constant over it (WPH15 Oct 2023).
- Ratio questions: for two isotopes with the same activity, $N_1/N_2 = \lambda_2/\lambda_1 = t_{1/2_1}/t_{1/2_2}$ (WPH15 Oct 2024).

Quick checks

QUICK CHECK

A radioactive sample is heated from 20 °C to 200 °C. What happens to its half-life?

- It decreases, because the nuclei have more energy.
- It increases, because the sample expands.
- It stays the same.
- It decreases at first and then returns to normal.

Decay is spontaneous: it isn't affected by temperature, pressure or chemistry. Ordinary heating changes the motion of whole atoms, which is nowhere near enough to affect what happens inside the nucleus.

QUICK CHECK

What fraction of a radioactive isotope remains undecayed after three half-lives?

- 1/3
- 1/6
- 1/8
- 1/9

Each half-life halves what is left: $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$. It never drops to zero after a whole number of half-lives.

QUICK CHECK

Why do alpha particles have such a short range in air?

- They are absorbed by the nitrogen nuclei in the air.
- They are strongly ionising, so they lose their energy in many interactions over a short distance.
- They have no charge, so they are not affected by the air.
- They are emitted with much less energy than beta particles.

Alpha particles are heavy, slow and have charge +2e, so they ionise almost every atom they pass. Each ionisation takes some energy, so their few MeV are used up within a few centimetres. (Alpha particles are usually emitted with *more* energy than beta particles, not less.)

QUICK CHECK

Which of these reactions releases energy?

- Splitting an iron-56 nucleus into two smaller nuclei
- Joining two hydrogen-2 nuclei to make helium
- Joining two uranium-235 nuclei together
- Splitting a helium-4 nucleus into two hydrogen-2 nuclei

Energy is released when the products have a higher binding energy per nucleon. Fusing hydrogen-2 (about 1.1 MeV per nucleon) into helium-4 (7.1 MeV per nucleon) moves steeply up the curve. The other three all move down the curve, so they would need energy to be supplied.

QUICK CHECK

Why must the fuel in a fusion reactor be at a very high temperature?

- So the nuclei can release their electrons and become neutral.
- So the strong force between nuclei becomes stronger.
- **So the nuclei have enough kinetic energy to overcome their electrostatic repulsion and get close enough to fuse.**
- So the fuel has a higher density.

Nuclei are positive and repel. They have to get within about 10^{-15} m for the short-range strong force to act, which needs enormous kinetic energy, and so an enormous temperature. (High density is needed too, but for a different reason: to make collisions frequent enough.)

QUICK CHECK

A graph of ln(corrected count rate) against time is a straight line with gradient −0.0231 s^{-1}. What is the half-life?

- 0.016 s
- **30 s**
- 43 s
- 0.0231 s

The gradient is $-\lambda$, so $\lambda = 0.0231\ \text{s}^{-1}$ and $t_{1/2} = \ln 2/\lambda = 0.693/0.0231 = 30\ \text{s}$. (43 s is $1/\lambda$, which is not the half-life.)

QUICK CHECK

Polonium-210 ($Z = 84$) decays by alpha emission. What is the product?

- $^{210}_{82}\text{Pb}$
- **$^{206}_{82}\text{Pb}$**
- $^{206}_{84}\text{Po}$
- $^{210}_{85}\text{At}$

Alpha emission removes $^{4}_{2}\alpha$: $A = 210 - 4 = 206$ and $Z = 84 - 2 = 82$, which is lead. Check: $210 = 206 + 4$ and $84 = 82 + 2$.